

CSI131 - DISCRETE STRUCTURES I

COURSE OUTLINE 2024/2025 - SEMESTER 1 DEPARTMENT OF COMPUTER SCIENCE UNIVERSITY OF BOTSWANA

INSTRUCTORS

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ADMIN INFORMATION

- Schedule : MW(F)-0900hrs, with Friday converted to 2 hours of Lab/Tutorial:
 - F 9-11am, 232-004, 232-011, 233-G06
 - F 11-13pm, 233-G03, 233-G11, 232-005
- Course website : www.cs.ub.bw/moodle - Assignments, course slides and announcements
- Consultations : MTW (0900hrs - 1500hrs)

GRADING

Table 1: assessment weights

Assessment	Weight (%)
Quiz 1	10%
Test 1	15%
Quiz 2	10%
Test 2	15%
Exam	50%

COURSE TEXTBOOK

- Prime textbook
 - Rosen Kenneth H. (2019). Eighth Edition. Discrete Mathematics and Its Applications. McGraw-Hill, New York - ISBN 978-1-259-67651-2.
- Supplementary textbooks/materials
 - Ensley D., and Crawley J., Discrete Mathematics: Mathematical Reasoning and Proof with Puzzles, Patterns, and Games , John Wiley & Sons, Inc., 2006.
 - Cupillari A., The Nuts and Bolts of Proofs, Academic Press, 2001.
 - online resources

IMPORTANT DATES

Table 2: Tentative test, quiz and assignment dates

Assessment	Date	Time	Venue
Quiz 1	TBA	TBA	TBA
Test 1	TBA	TBA	TBA
Short break	No break this semester		
Quiz 2	TBA	TBA	TBA
Test 2	TBA	TBA	TBA
Exam	TBA	TBA	TBA

COURSE OVERVIEW

What is this course all about

- **Discrete Mathematics** Is the part of mathematics devoted to the study of discrete objects. Used when objects are counted, when relationships between finite sets are studied, and when processes involving a finite number of steps are analyzed.
- **Discrete** Means consisting of distinct or unconnected elements.
- **The kind of problems solved using discrete mathematics include:**
 - What is the probability of winning a lottery?
 - Is there a link between two computers in a network?
 - How many ways are there to choose a valid password on a computer system?

WHICH DISCRETE STRUCTURES WILL BE COVERED? - SOME IN DISCRETE STRUCTURES II

- | | | | |
|--------------------|--------------|----------------|------------------|
| • Propositions | • Algorithms | • Permutations | • Logic Circuits |
| • Predicates | • Integers | • Combinations | • Automata |
| • Sets | • Summations | • Relations | |
| • Functions | • Sequences | • Graphs | |
| • Orders of growth | • Strings | • Trees | |

COURSE OBJECTIVES AND LEARNING OUTCOMES

- The main purpose of this course is to introduce mathematical concepts and ideas in discrete mathematics which are required in all of computer Science. After completion of the course, students should be able to:
 - read and understand basic mathematical proofs
 - state and prove non-trivial facts
 - recite mathematical ideas and notation used in Computer Science
 - describe the importance of algorithms in solving problems
 - demonstrate and apply basic counting principles

COURSE SYNOPSIS (TAKEN FROM DEPARTMENTAL CATALOG)

Functions, relations and sets; Propositional logic and predicate calculus; Algorithms as effective processes; Proof techniques; Algorithms, the integers and matrices; Basic counting; Boolean algebra.

COURSE OUTLINE AND RELEVANT CHAPTERS IN COURSE TEXTBOOK)

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|---------------------------------|--------------------------|
| 1. logic (1.1 – 1.5) | 7. integers (3.4 – 3.5) |
| 2. proofs (1.6 – 1.7) | 8. matrices (3.8) |
| 3. set theory (2.1 – 2.2) | 9. induction (4.1 – 4.2) |
| 4. functions (2.3) | 10. recursion (4.3) |
| 5. sequences & summations (2.4) | 11. counting (5.1 – 5.3) |
| 6. algorithms (3.1 – 3.3) | |

Tentative Syllabus

Week	Topic	Duration	Content
1 (11-16 Aug)	Propositional Logic	2 hours of lectures. Self study	Propositions, Negation, Disjunction, Conjunction, Implication, Biconditional.
2 (18-22 Aug)	Truth Tables of Compound Propositions	2 hours of lectures. 2 hours of tutorial. Self study	Truth Tables for Compound Propositions. Translating Propositional Logic Statements. Solving Logic Puzzles. Logic Circuits.
3 (25-29 Aug)	Propositional Equivalences	2 hours of lectures. 2 hours of tutorial. Self study	Logical Equivalences with Truth Tables. Key Logical Equivalences including De Morgan's Laws. Constructing New Logical Equivalences.
4 (1-4 Sept)	Predicates and Quantifiers	2 hours of lectures. 2 hours of tutorial. Self study	Predicate Logic. Quantifiers. Negating and Translating with Quantifiers. Translating with Nested Quantifiers.
5 (8-12 Sept)	Rules of Inference	2 hours of lectures. 2 hours of tutorial. Self study	Rules of Inference for Propositional Logic. Rules of Inference for Quantified Statements.
6(15-19 Sept)	Introduction to Proofs	2 hours of lectures. 2 hours of tutorial. Self study	Direct Proof. Proof by Contraposition. Proof by Contradiction.
7(22-26 Sept)	Set Theory	2 hours of lectures. 2 hours of tutorial. Self study	Introduction to Sets. Set Relationships. Operations on Sets.
	BREAK CANCELLED	BREAK CANCELLED	BREAK CANCELLED
8 (29 Sep-3 Oct)	Set Identities	2 hours of lectures. 2 hours of tutorial. Self study	Set Identities. Proving Set Identities.
9 (6-10 Oct)	Functions	2 hours of lectures. 2 hours of tutorial. Self study	One-to-one and Onto Functions. Inverse Functions. Composition, Useful Functions. Graphs.
10(13-17 Oct)	Sequences and Summations	2 hours of lectures. 2 hours of tutorial. Self study	Introduction to Sequences. Recurrence Relations. Summations and Sigma Notation. Summation Properties and Formulas.
11(20-24 Oct)	Mathematical Induction	2 hours of lectures. 2 hours of tutorial. Self study	Proof by Induction
12(27-31 Oct)	Recursion	2 hours of lectures. 2 hours of tutorial. Self study	Recursively Defined Functions. Recursive Algorithms.
13 (3-7 Nov)	Algorithms	2 hours of lectures. 2 hours of tutorial. Self study	Introduction to Algorithms and Pseudocode. Searching Algorithms. Sorting Algorithms.
14(10-14 Nov)	EXAMS	EXAMS	EXAMS

ACADEMIC DISHONESTY FLYER FROM THE "OFFICE OF THE DEPUTY
VICE CHANCELLOR, STUDENT AFFAIRS"

ACADEMIC DISHONESTY

- harms the good reputation of our University
- affects your personal integrity
- is a barrier to the acquisition of the knowledge, skills and attitudes you would have acquired at the end of your study
- decreases your value in the eyes of employers
- has a heavy penalty including expulsion from UB

DO NOT CHEAT OR CONDONE CHEATING BY OTHERS